杉山　素直

履歴書, 最終更新日: July 27, 2026

連絡先

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GitHub	`https://github.com/git-sunao`
Researchmap	`https://researchmap.jp/sunaosugiyama`

研究分野

理論および観測的宇宙論:
宇宙の大規模構造、重力レンズ（弱レンズ、マイクロレンズ）、原始ブラックホール

COLLABORATIONS

Subaru HSC weak lensing working group, member (2021-present, **co-chair** since 2024Nov)
Dark Enrgy Survey (DES), member (2024-present), LSST member(2024-present), Roman HLIS (2024-present)

職歴

現在 **特任研究員**, 2026 年 2 月 – 現在
日本, カブリ数物連携宇宙研究機構, 千葉

過去 **ポスドク研究員**, 2023 年 9 月 – 2026 年 1 月
アメリカ合衆国, ペンシルベニア大学, フィラデルフィア
受入教員: Bhuvnesh Jain

JSPS 海外特別研究員, 2023 年 9 月 – 2025 年 8 月
アメリカ合衆国, ペンシルベニア大学, フィラデルフィア

特任研究員, 2023 年 4 月 – 2023 年 8 月
日本, カブリ数物連携宇宙研究機構, 千葉
受入教員: 高田昌広, funded partially by beyond AI

日本学術振興会特別研究員（DC2）, 2021 年 4 月 – 2023 年 3 月
日本, カブリ数物連携宇宙研究機構, 千葉

学歴 **東京大学, 東京, 日本**, 2020 年 4 月 – 2023 年 3 月
物理学専攻, 博士課程
論文題目: *“Joint cosmology analyses using gravitational weak lensing data from Subaru Hyper Suprime-Cam”*
指導教員: 高田昌広

東京大学, 東京, 日本, 2018 年 4 月 – 2020 年 3 月
物理学専攻, 修士
論文: *“Validation of cosmological analysis based on perturbation theory for wide-field galaxy survey”*
指導教員: 高田昌広

東京大学, 東京, 日本, 2014 年 4 月 – 2018 年 3 月
物理学専攻, 学士

獲得研究資金 および 受賞

Grant-in-Aid for JSPS Research Fellows (DC2), Japan Society for the Promotion of Science, Apr. 2021 – Mar. 2023

理学系研究科奨励賞 (博士課程), 東京大学, 理学系, 2023 年 3 月

WINGS IGPEES, コース修了, Sep. 2018 – Mar. 2023

教育

Collaborative coding: git and github, CD3 symposium 2023, Kavli IPMU

Coadvised students: Rafael C. H. Gomes (a PhD student at UPenn since 2023), Kyle Miller (a visiting student at UPenn, since 2024), Noriaki Nakasawa (a master student at the University of Nagoya, 2022)

活動

学会	日本天文学会 (ASJ), 2018 年 – 現在 日本物理学会 (JPS), 2022 年 – 現在
企画・運営	IPMU ランチセミナー (共同オーガナイザー), 2019 年 – 2021 年 HSC 弱重力レンズミニワークショップ主催, 2022 年 8 月 Sesto 2025 - Tracing Cosmic Evolution with Galaxy Clusters V (SOC), 2025
レフェリー	International Journal of Modern Physics D The Astrophysical Journal Astronomy & Astrophysics Journal of Cosmology and Astroparticle Physics
コンピューティング	開発コード: `fft-extended-source`, `fastnc`, `dark emulator` (Dark Quest Project の一部) C、C++、Python、HSC パイプライン（画像解析用）を使用できます
採択された観測	Definitive search for PBH dark matter in the multiverse cosmology with HSC の **PI** Survey of M31 eclipsing binaries: Toward a 1% distance measurement の co-I

アウトリーチ, メディア協力

NHK コズミック　フロント　「原始ブラックホール　宇宙創成のマスターキー」 出演, 2021 年

Quanta Magazine on *Clashing Cosmic Numbers Challenge Our Best Theory of the Universe*, インタビュー, 2024 年

朝日新聞, 宇宙の標準理論にほころび？　暗黒物質の精密な「地図」で解析, インタビュー, 2024 年

Interviewed by Jonathan O'Callaghan for "Is This a Moon-Sized Primordial Black Hole Adrift in the Milky Way?" *Scientific American*, 2026.

出版/発表論文

最新の論文リストは ADS を参照ください。

* = 著者リストアルファベット順

筆頭著者または主要な貢献をした査読付論文

1. **Sugiyama, Sunao**, R. C. H. Gomes, and M. Jarvis. Fast modeling of the shear three-point correlation function. Phys. Rev. D, 113(12):123548, June 2026:123548
2. **Sugiyama, Sunao**, M. Takada, N. Yasuda, and N. Tominaga. Microlensing constraints on Primordial Black Hole abundance with Subaru Hyper Suprime-Cam observations of Andromeda. *arXiv e-prints*, arXiv:2602.05840, February 2026:arXiv:2602.05840
3. R. C. H. Gomes, K. Miller, **Sugiyama, Sunao**, et al. Tidal alignment and tidal torquing modeling for the cosmic shear three-point correlation function and mass aperture skewness. *arXiv e-prints*, arXiv:2601.09133, January 2026:arXiv:2601.09133
4. R. C. H. Gomes, **Sugiyama, S.**, B. Jain, et al. Cosmology with second- and third-order shear statistics for the Dark Energy Survey: Methods and simulated analysis. Phys. Rev. D, 112(12):123514, December 2025:123514
5. **Sugiyama, Sunao** and M. Park. Data compression with noise suppression for inference under noisy covariance. Phys. Rev. D, 112(12):123505, December 2025:123505
6. R. C. H. Gomes, **Sugiyama, S.**, B. Jain, et al. Dark Energy Survey Year 3 Results: Cosmological constraints from second- and third-order shear statistics. Phys. Rev. D, 112(12):123515, December 2025:123515
7. **Sugiyama, Sunao**, R. C. H. Gomes, and B. Jain. Cosmology from a joint analysis of second- and third-order shear statistics with Subaru Hyper Suprime-Cam Year 3 data. Phys. Rev. D, 112(12):123545, December 2025:123545
8. X. Li, T. Zhang, **Sugiyama, Sunao**, et al. Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear two-point correlation functions. Phys. Rev. D, 108(12):123518, December 2023:123518
9. H. Miyatake, **Sugiyama, Sunao**, M. Takada, et al. Hyper Suprime-Cam Year 3 results: Cosmology from galaxy clustering and weak lensing with HSC and SDSS using the emulator based halo model. Phys. Rev. D, 108(12):123517, December 2023:123517
10. **Sugiyama, Sunao**, H. Miyatake, S. More, et al. Hyper Suprime-Cam Year 3 results: Cosmology from galaxy clustering and weak lensing with HSC and SDSS using the minimal bias model. Phys. Rev. D, 108(12):123521, December 2023:123521
11. R. Dalal, X. Li, A. Nicola, et al. Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear power spectra. Phys. Rev. D, 108(12):123519, December 2023:123519
12. S. More, **Sugiyama, Sunao**, H. Miyatake, et al. Hyper Suprime-Cam Year 3 results: Measurements of clustering of SDSS-BOSS galaxies, galaxy-galaxy lensing, and cosmic shear. Phys. Rev. D, 108(12):123520, December 2023:123520
13. **Sugiyama, Sunao**, M. Takada, and A. Kusenko. Possible evidence of axion stars in HSC and OGLE microlensing events. *Physics Letters B*, 840:137891, May 2023:137891
14. H. Miyatake, **Sugiyama, Sunao**, M. Takada, et al. Cosmological inference from an emulator based halo model. II. Joint analysis of galaxy-galaxy weak lensing and galaxy clustering from HSC-Y1 and SDSS. Phys. Rev. D, 106(8):083520, October 2022:083520
15. H. Miyatake, Y. Kobayashi, M. Takada, et al. Cosmological inference from an emulator based halo model. I. Validation tests with HSC and SDSS mock catalogs. Phys. Rev. D, 106(8):083519, October 2022:083519
16. **Sugiyama, Sunao**. Fast Fourier Transformation Based Evaluation of Microlensing Magnification with Extended Source. ApJ, 937(2):63, October 2022:63
17. **Sugiyama, Sunao**, M. Takada, H. Miyatake, et al. HSC Year 1 cosmology results with the minimal bias method: HSC ×BOSS galaxy-galaxy weak lensing and BOSS galaxy clustering. Phys. Rev. D, 105(12):123537, June 2022:123537

18. **Sugiyama, Sunao**, V. Takhistov, E. Vitagliano, et al. Testing stochastic gravitational wave signals from primordial black holes with optical telescopes. *Physics Letters B*, 814:136097, March 2021:136097
19. *A. Kusenko, M. Sasaki, **Sugiyama, Sunao**, et al. Exploring Primordial Black Holes from the Multiverse with Optical Telescopes. Phys. Rev. Lett., 125(18):181304, October 2020:181304
20. **Sugiyama, Sunao**, M. Takada, Y. Kobayashi, et al. Validating a minimal galaxy bias method for cosmological parameter inference using HSC-SDSS mock catalogs. Phys. Rev. D, 102(8):083520, October 2020:083520
21. **Sugiyama, Sunao**, T. Kurita, and M. Takada. On the wave optics effect on primordial black hole constraints from optical microlensing search. MNRAS, 493(3):3632–3641, April 2020:3632–3641
22. H. Niikura, M. Takada, N. Yasuda, et al. Microlensing constraints on primordial black holes with Subaru/HSC Andromeda observations. *Nature Astronomy*, 3:524–534, April 2019:524–534

その他の共著論文

23. T. Zhang, **Sugiyama, Sunao**, S. More, et al. Modeling galaxy clustering and tomographic galaxy-galaxy lensing with HSC Y3 and SDSS data using the point-mass correction model and redshift self-calibration. Phys. Rev. D, 113(10):103529, May 2026:103529
24. T. Zhang, X. Li, **Sugiyama, Sunao**, et al. Cosmology and source redshift constraints from galaxy clustering and tomographic weak lensing with HSC Y3 and SDSS using the point-mass correction model. Phys. Rev. D, 113(10):103530, May 2026:103530
25. D. Rana, S. More, H. Miyatake, et al. Hyper Suprime-Cam Y3 results: Photo-z bias calibration with lensing shear ratios and cosmological constraints from cosmic shear. Phys. Rev. D, 113(6):063534, March 2026:063534
26. R. Terasawa, M. Takada, T. Kurita, and **Sugiyama, Sunao**. Late-time suppression of structure growth as a solution for the S8 tension. Phys. Rev. D, 112(8):083556, October 2025:083556
27. R. Terasawa, M. Takada, **Sugiyama, Sunao**, and T. Kurita. Testing small-scale modifications in the primordial power spectrum with Subaru HSC cosmic shear, primary CMB, and CMB lensing data. Phys. Rev. D, 112(4):043514, August 2025:043514
28. R. Terasawa, X. Li, M. Takada, et al. Exploring the baryonic effect signature in the Hyper Suprime-Cam Year 3 cosmic shear two-point correlations on small scales: The S8 tension remains present. Phys. Rev. D, 111(6):063509, March 2025:063509
29. K.-F. Chen, I.-N. Chiu, M. Oguri, et al. Weak-Lensing Shear-Selected Galaxy Clusters from the Hyper Suprime-Cam Subaru Strategic Program: I. Cluster Catalog, Selection Function and Mass-Observable Relation. *The Open Journal of Astrophysics*, 8:2, January 2025:2
30. T. Sunayama, H. Miyatake, **Sugiyama, Sunao**, et al. Optical cluster cosmology with SDSS redMaPPer clusters and HSC-Y3 lensing measurements. Phys. Rev. D, 110(8):083511, October 2024:083511
31. I.-N. Chiu, K.-F. Chen, M. Oguri, et al. Weak-Lensing Shear-Selected Galaxy Clusters from the Hyper Suprime-Cam Subaru Strategic Program: II. Cosmological Constraints from the Cluster Abundance. *The Open Journal of Astrophysics*, 7:90, October 2024:90
32. J. Shi, T. Sunayama, T. Kurita, et al. The intrinsic alignment of galaxy clusters and impact of projection effects. MNRAS, 528(2):1487–1499, February 2024:1487–1499
33. T. Zhang, X. Li, R. Dalal, et al. A general framework for removing point-spread function additive systematics in cosmological weak lensing analysis. MNRAS, 525(2):2441–2471, October 2023:2441–2471
34. Y. Park, T. Sunayama, M. Takada, et al. Cluster cosmology with anisotropic boosts: validation of a novel forward modelling analysis and application on SDSS redMaPPer clusters. MNRAS, 518(4):5171–5189, February 2023:5171–5189

他の記事

1. S. Sugiyama, M. Takada, and H. Miyatake. Weak lensing cosmology with subaru hsc data. *ASJ EUREKA*, 117(1):304–314, May 2024:304–314

講演

全 50 件のうち 20 件のトークを選出しました。全リストはこちらをご覧ください。

1. **Probing Primordial Black Hole Dark Matter with Subaru HSC Microlensing Observations of M31**, IPNS Joint Experimental-Theoretica Cosmology Seminar, 2026, Jun., *Oral* (**Invited Talk**)
2. **Searching for Primordial Black Holes with Microlensing Data from Subaru HSC**, ASIAA colloquium, 2026, Apr., *Oral* (**Invited Talk**)
3. **Observational Search for Primordial Black Holes Using Subaru/HSC Microlensing**, Elucidating the Universe's Creation with Neutrinos based on International Science Collaborations, 2026, Mar., *Oral* (**Invited Talk**)
4. **Observational search of PBH**, Dark matter and black holes, 2025, Dec., *Oral* (**Invited Talk**)
5. **Data Compression with Noise Suppression for Inference under Noisy Covariance**, Roman F2F meeting, 2025, Nov., *Oral*
6. **Cosmology from a joint analysis of second and third order shear statistics**, KICP seminar, 2025, Oct., *Oral*
7. **Cosmology with third-order shear statistics Applications to HSC and DES**, Beyond-two-point Statistics Meet Survey Systematics, 2025, Sep., *Oral* (**Invited Talk**)
8. **Exploring Primordial Black Hole with Microlensing Data: Updates on Analysis Pipeline**, Focus week on primordial black holes 2024, 2024, Nov., *Oral* (**Invited Talk**)
9. **Exploring Primordial Black Hole with Microlensing Data**, Pacific conference, 2024, Aug., *Oral* (**Invited Talk**)
10. **Cosmology from Subaru HSC weak lensing Year 3 data**, MIfA colloquium, 2024, May., *Oral* (**Invited Talk**)
11. **Cosmology from weak lensing three-point correlation function**, astro/cosmo seminar at CMU, 2024, Feb., *Oral*
12. **Cosmology from Subaru HSC weak lensing Year 3 data**, Subaru Users Meeting FY2023, 2024, Jan., *Oral*
13. **HSC Y3 weak lensing cosmology results**, CosmoPalooza, 2023, Oct., *Oral*
14. **HSC Year 3 Weak Lensing Cosmology Results**, HSC webinar, 2023, Apr., *Oral*
15. **HSC Y3 cosmology results**, CMB x LSS, 2023, Apr., *Oral* (**Invited Talk**)
16. **Collaborative coding: git and github**, CD3 Opening Symposium, 2023, Apr., *Oral*
17. **Revealing the nature of dark matter with gravitational lensing: weak and microlensing**, Colloqium at Osaka theoretical astrophysics group, 2022, Jul., *Oral* (**Invited Talk**)
18. **Exploring Primordial black hole with microlensing observation of Andromeda galaxy**, Subaru Users Meeting 2021, 2022, Jan., *Oral*
19. **すばる HSC と SDSS データの銀河弱重力レンズとクラスタリングの大スケール信号を用いた宇宙論統合解析**, 天文学会 2021 年秋季年会, 2021, Sep., *Oral*
20. **Testing stochastic gravitational wave signals by PBH microlensing**, 4th KEK-PH + KEK-Cosmo Joint Lectures and Workshop on "Gravitational Wave", 2020, Nov., *Oral* (**Invited Talk**)

プレスリリース

原始ブラックホールと多元宇宙が予言するダークマターの探索, IPMU, 2020 Dec

ダークマターを見る！ – HSC 国際チームが宇宙の標準理論を検証, IPMU, 2024 Apr